Idealised ruler and collapsing compass: Euclidean geometry

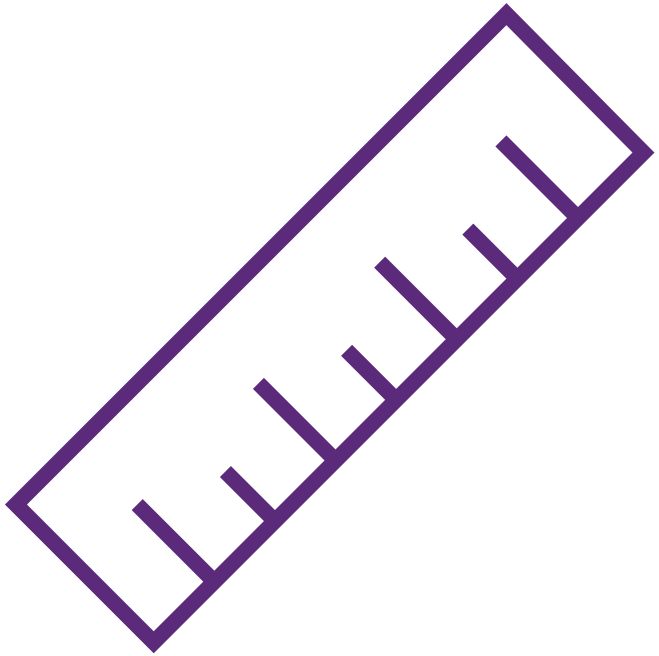
Veronika Lohmanova

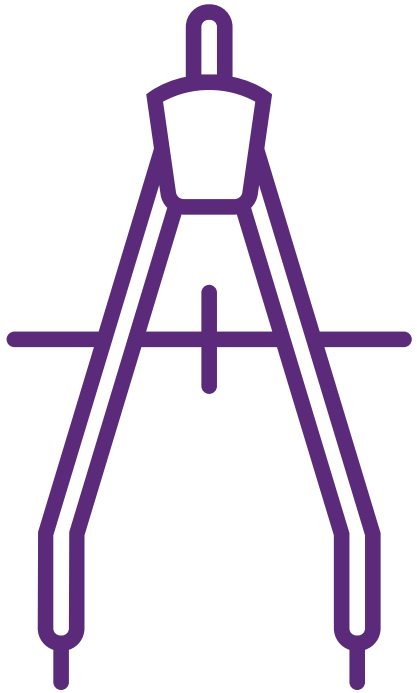
Manchester Interdisciplinary Mathematics Undergraduate Conference (MIMUC) 2025

Agenda

- Problem setting
- Method of geometric places
- Building toolset
- Notation
- Answering the question
- Why?

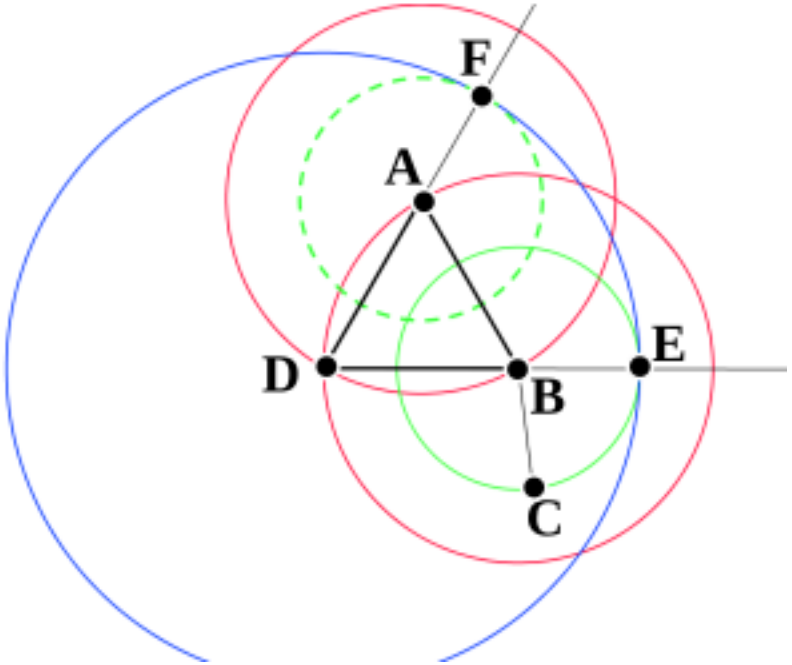
Problem setting

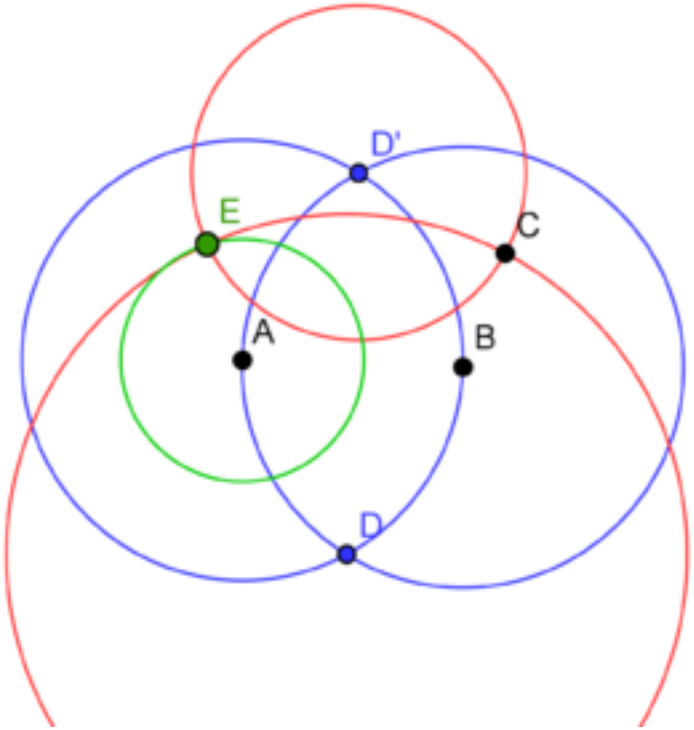




Problem setting

Mohr–Mascheroni theorem

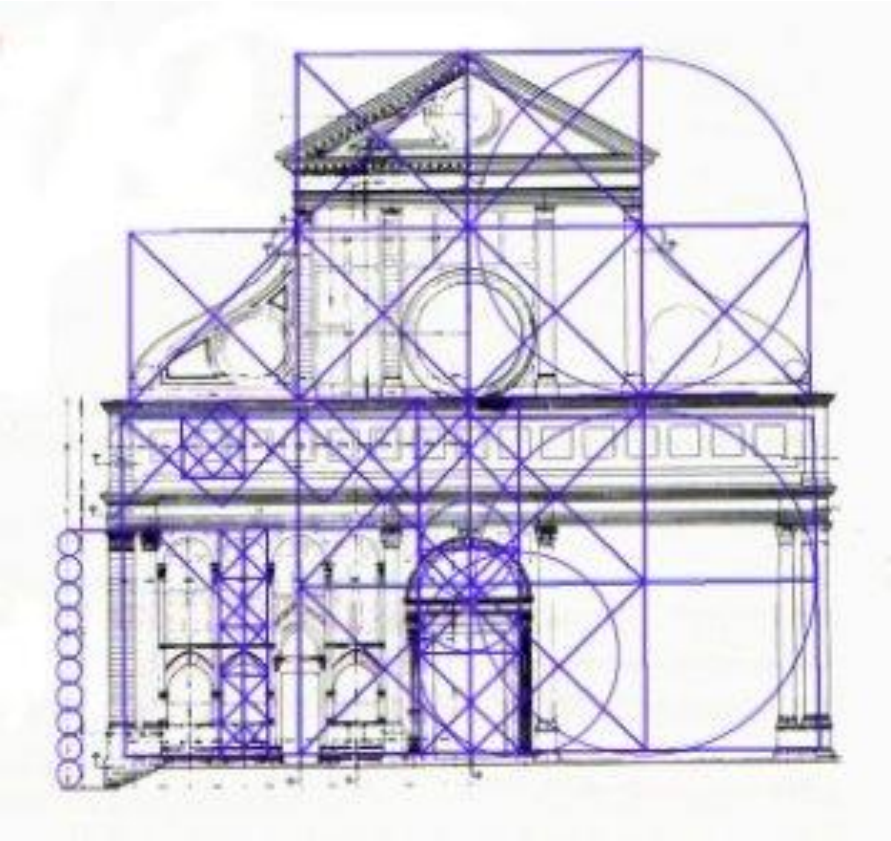




Compass equivalence theorem

Source: Wikipedia: Compass Equivalence theorem

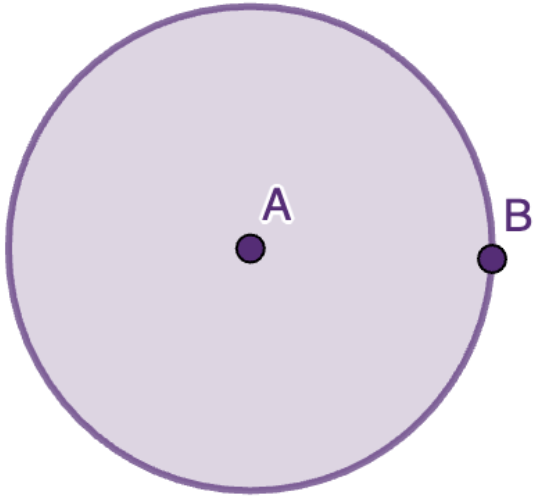
Why?

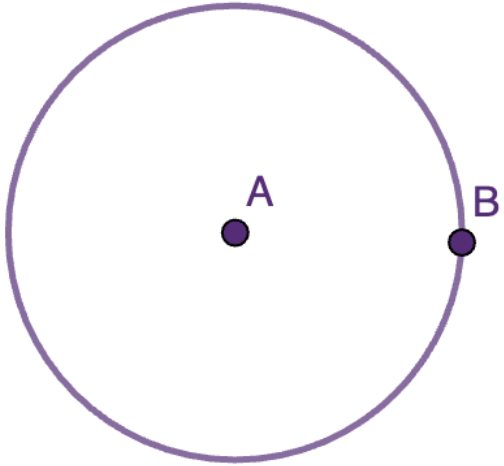


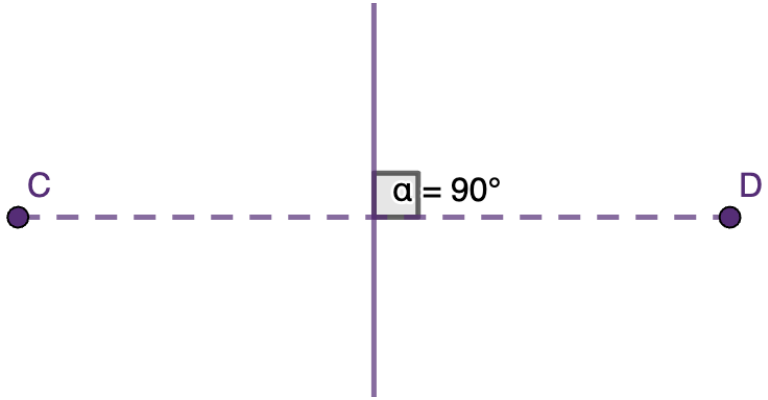
Source: Toeyyboonchit: Response to Renaissance Arcitecture. Geometric shape?

Method of geometric places

- Locus







Method of geometric places

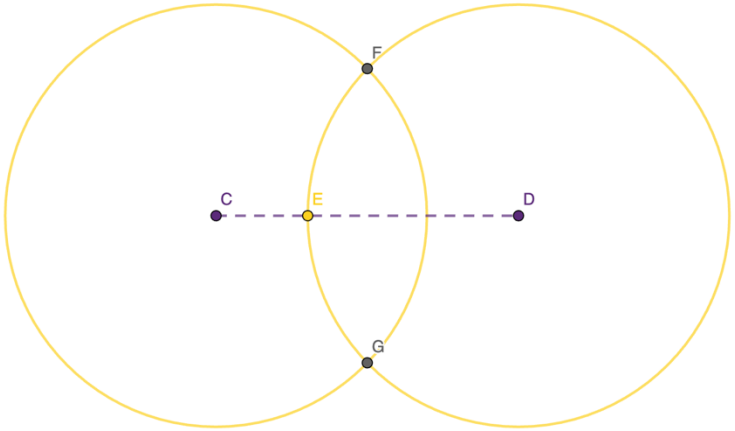
- Intersection of geometric places (the pattern of two loci)

"First, reduce the problem to the construction of ONE point.

Then, split the condition into TWO parts so that each part yields a locus for the unknown point"

Method of geometric places

- Intersection of geometric places (the pattern of two loci)



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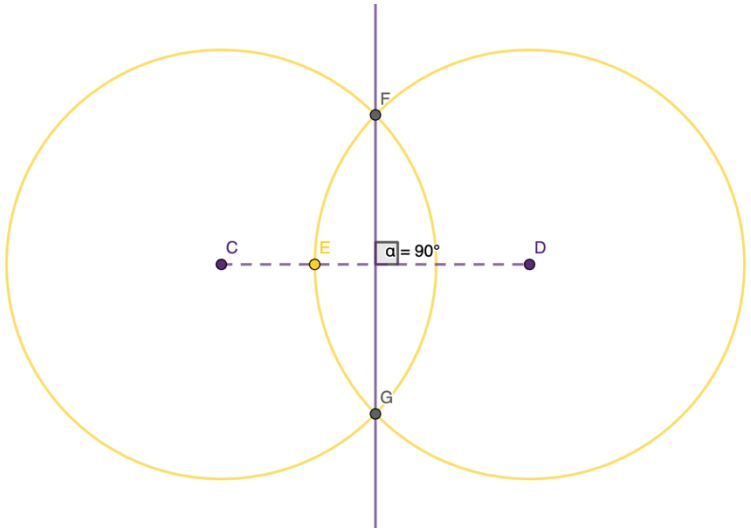

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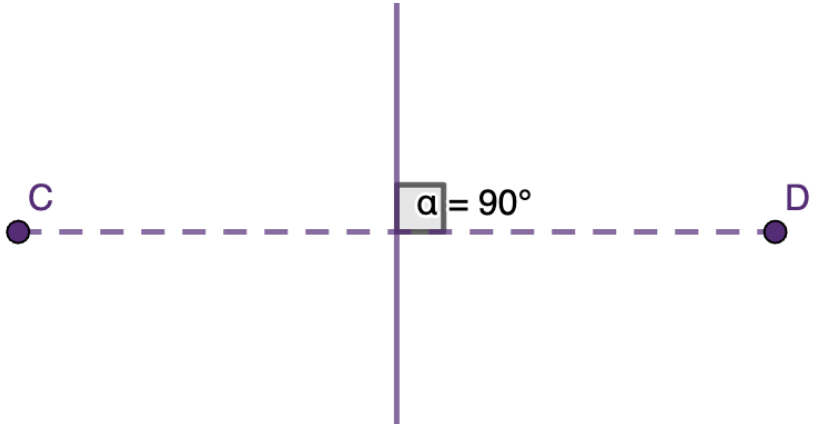

Example

- Given two parallel lines and a point between them. Draw a circle that is tangent to both given lines and passes through the given point

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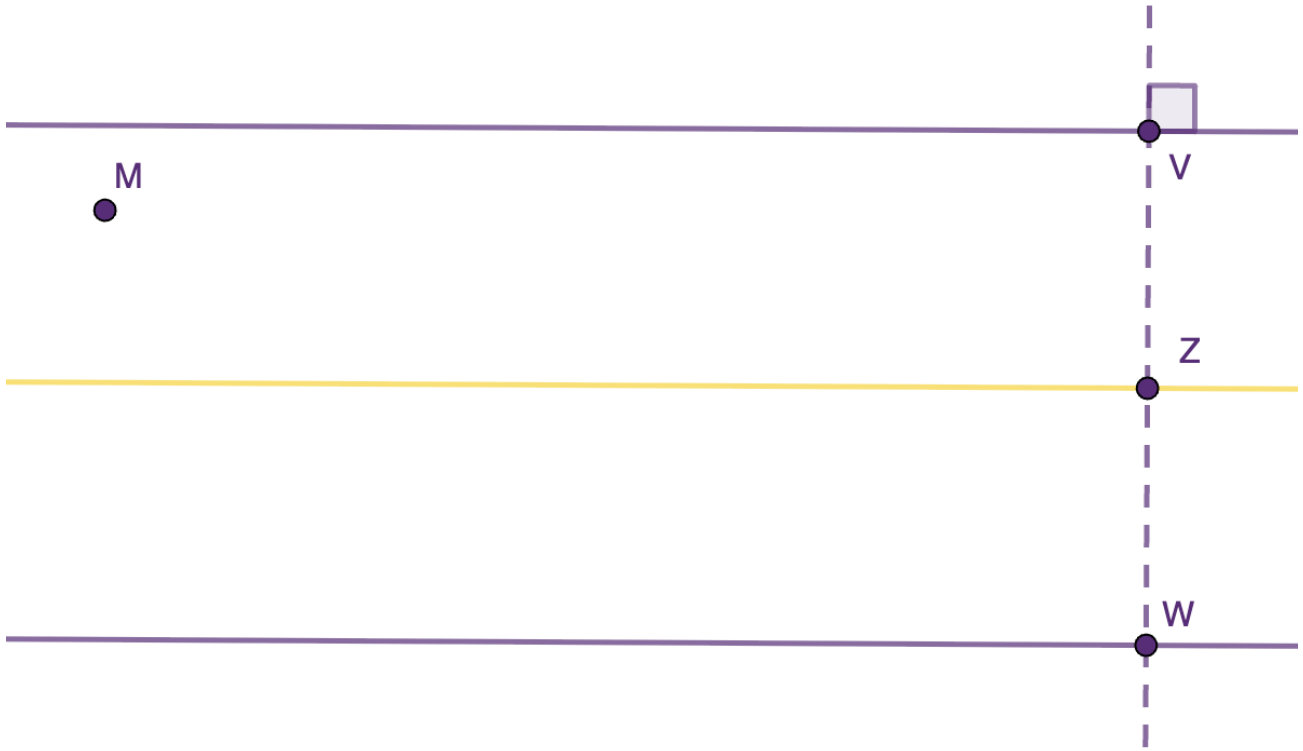

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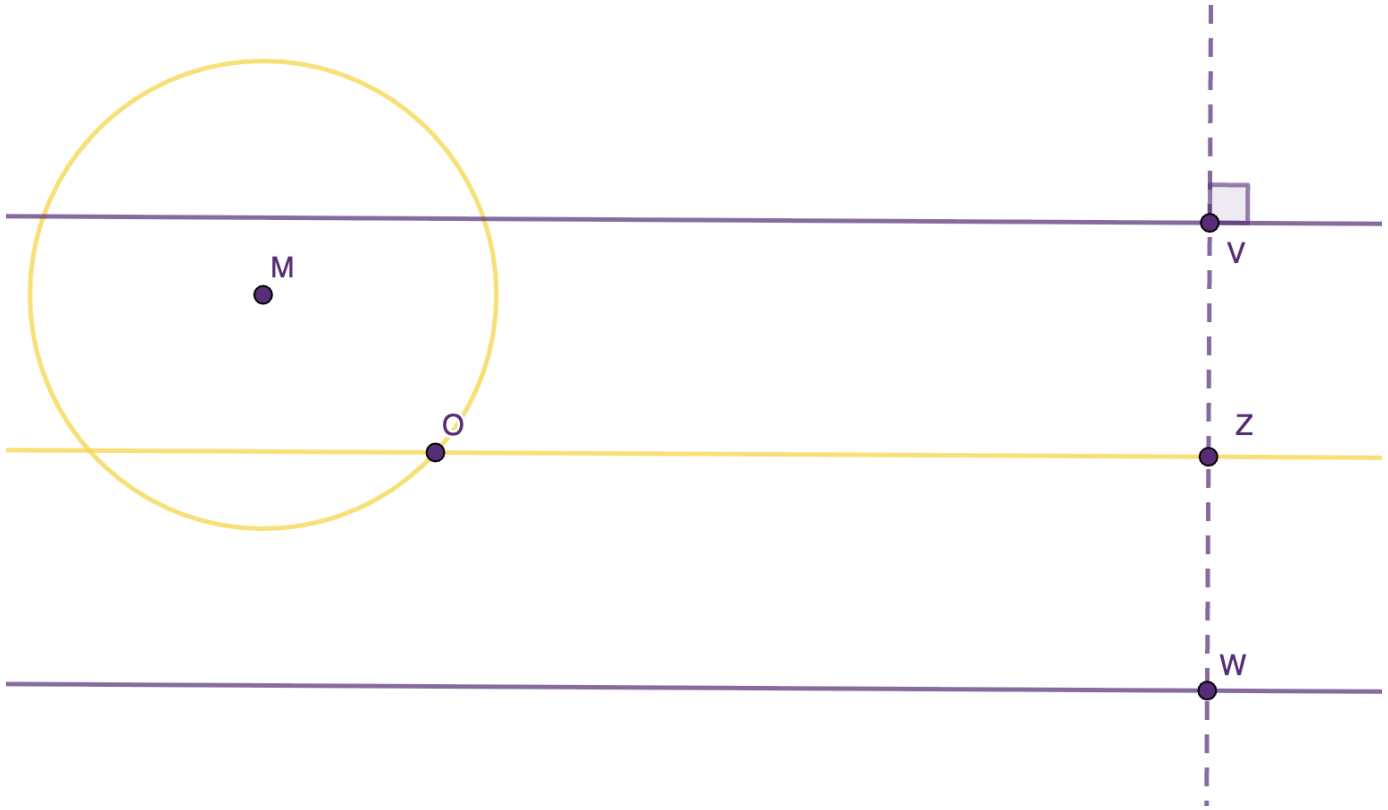

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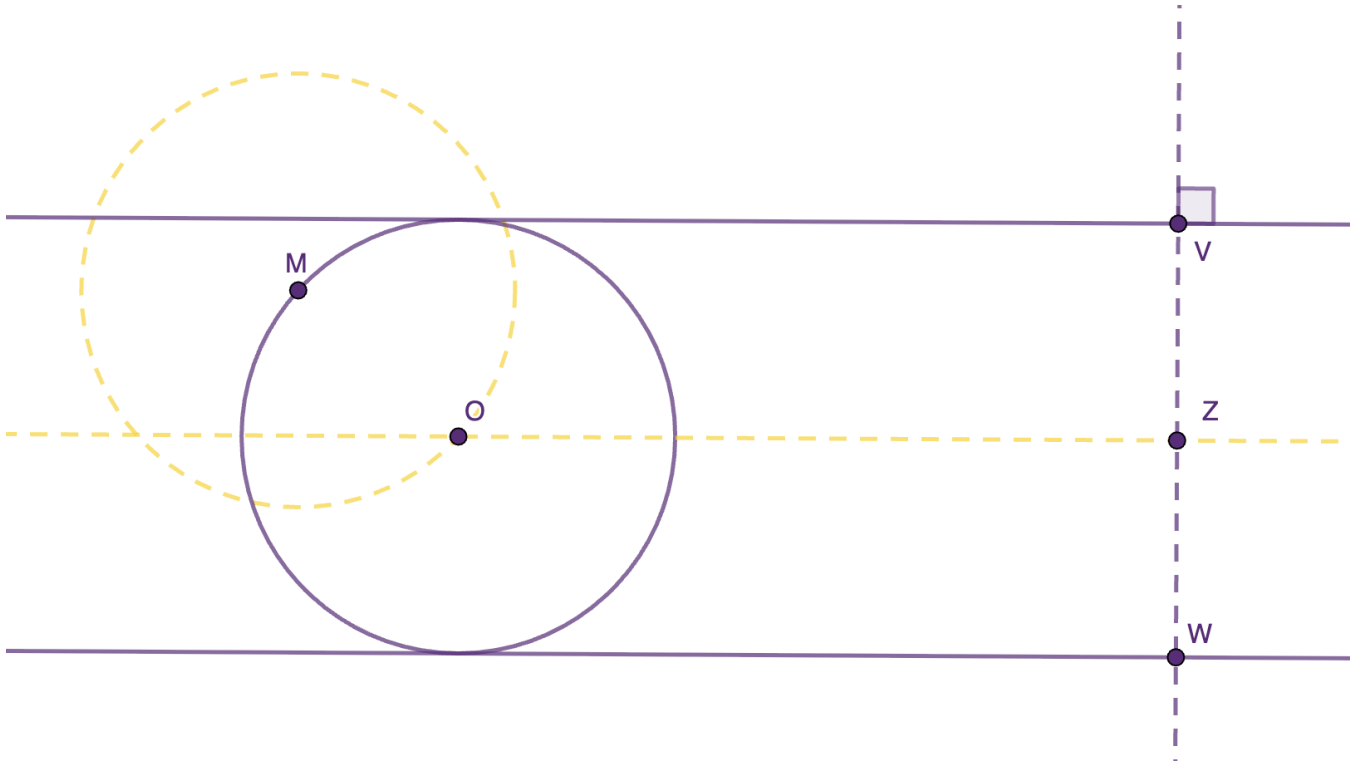
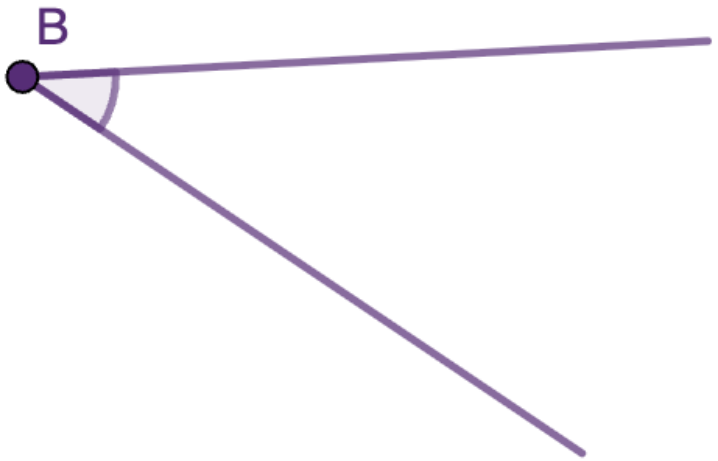


Building the toolset

- Draw a circle with the given radius ✅
- Draw a segment on the line with a given length ✅
- Draw a perpendicular bisector ✅
- Find midpoint of given segment ✅
- ???
- Replicate given angle

Building the toolset: replicating angles



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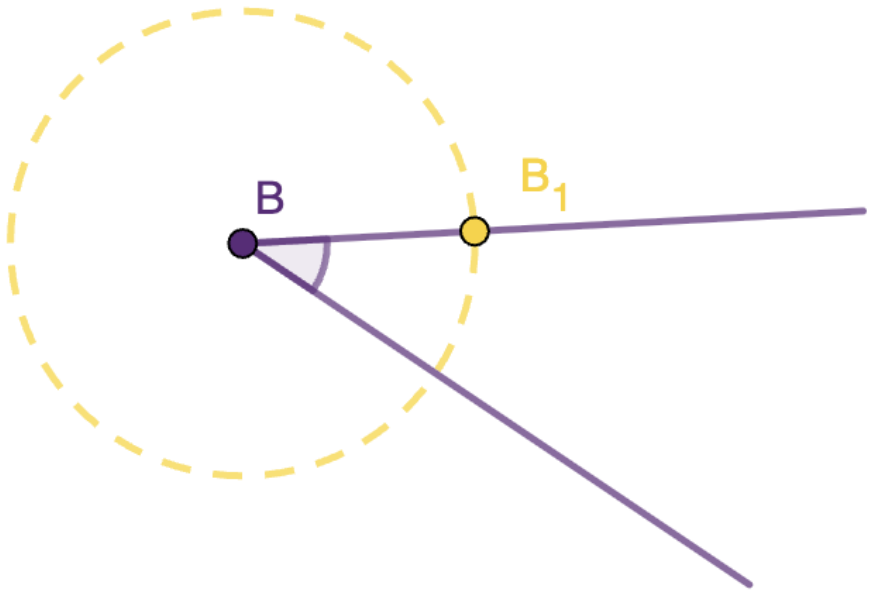

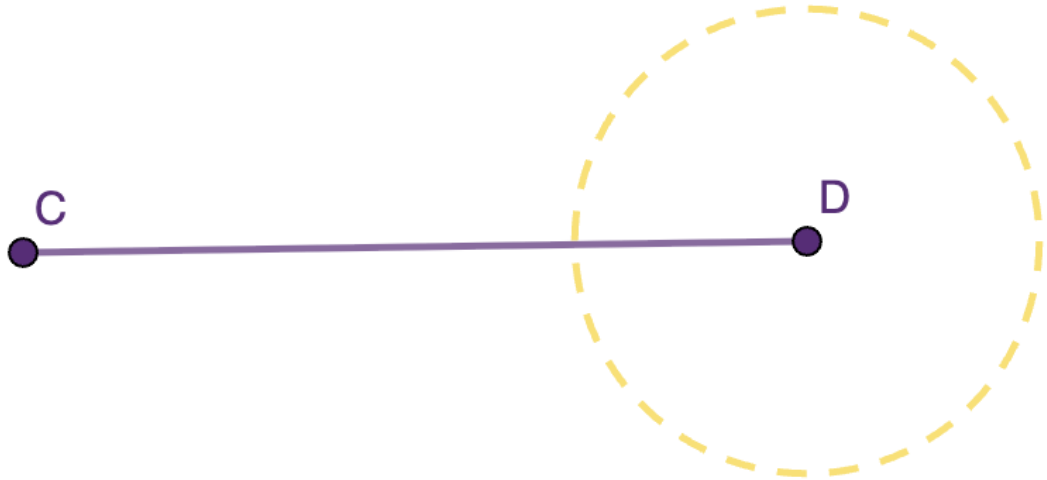

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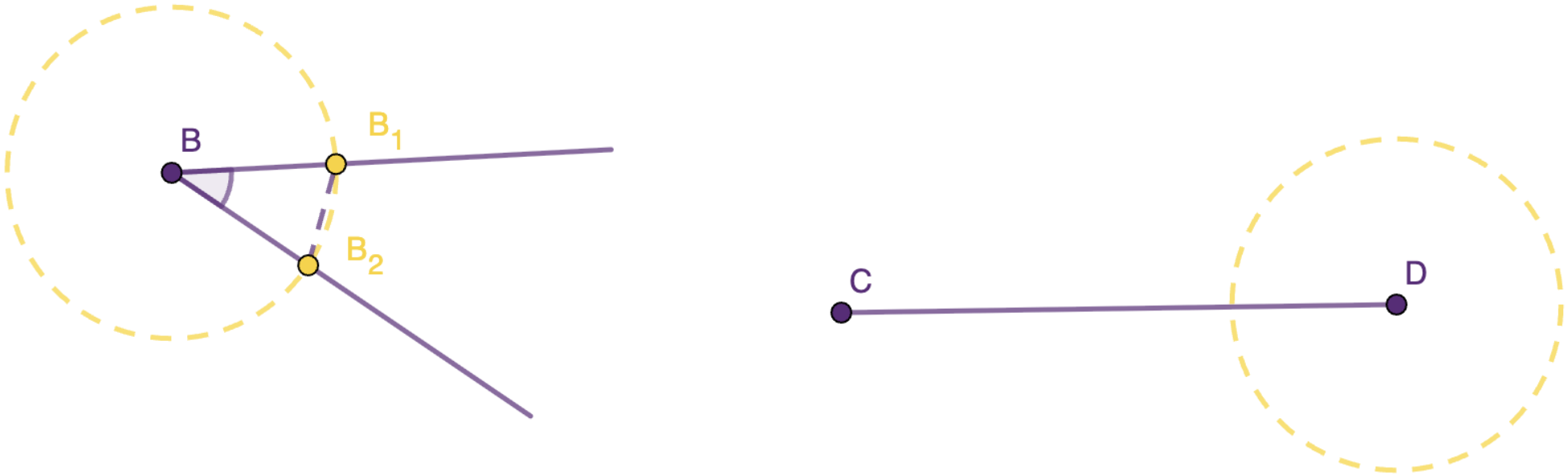

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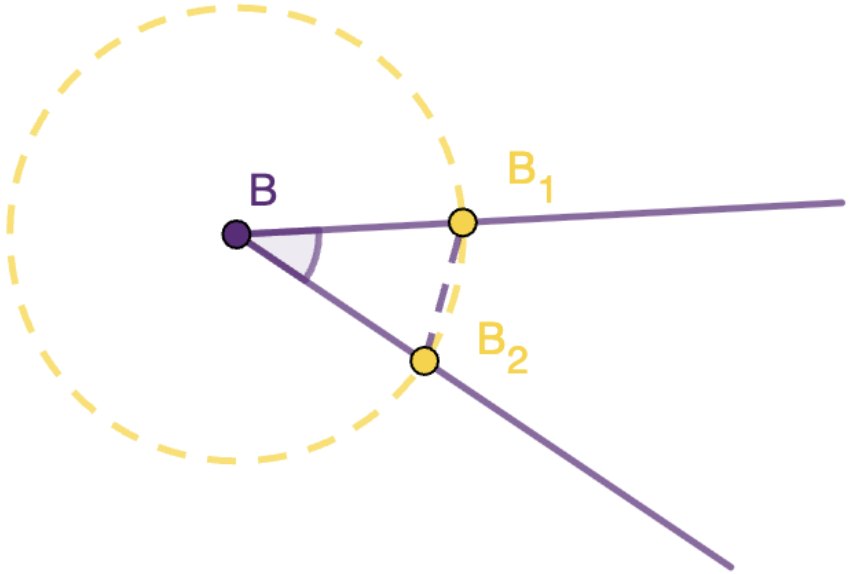

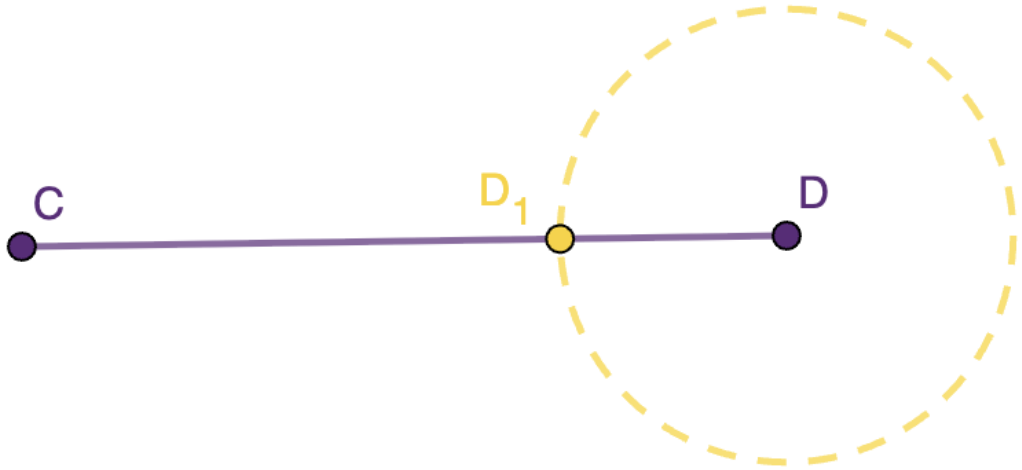

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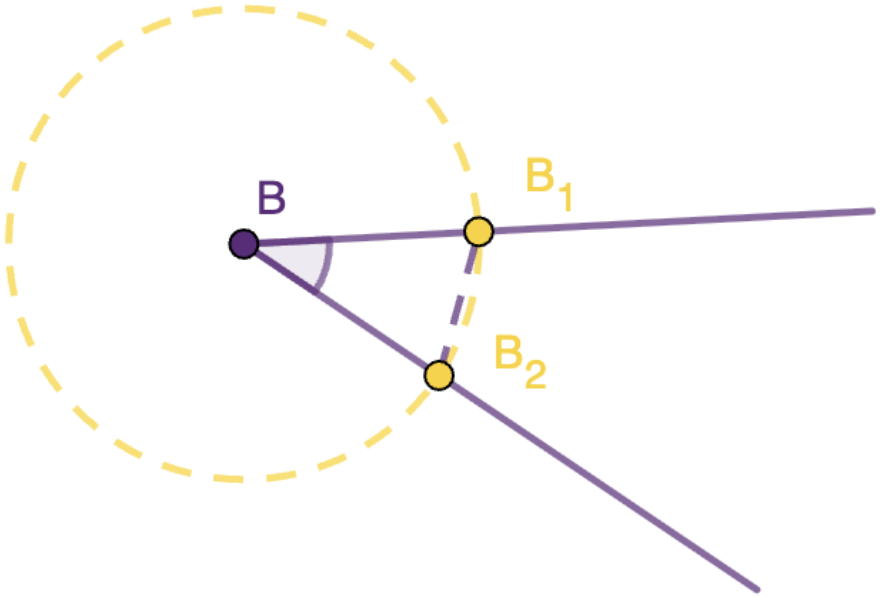

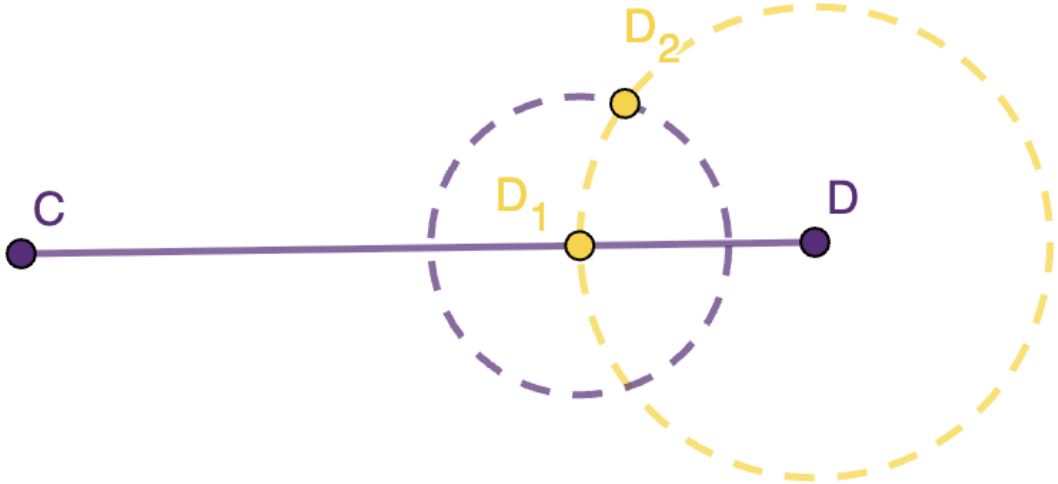

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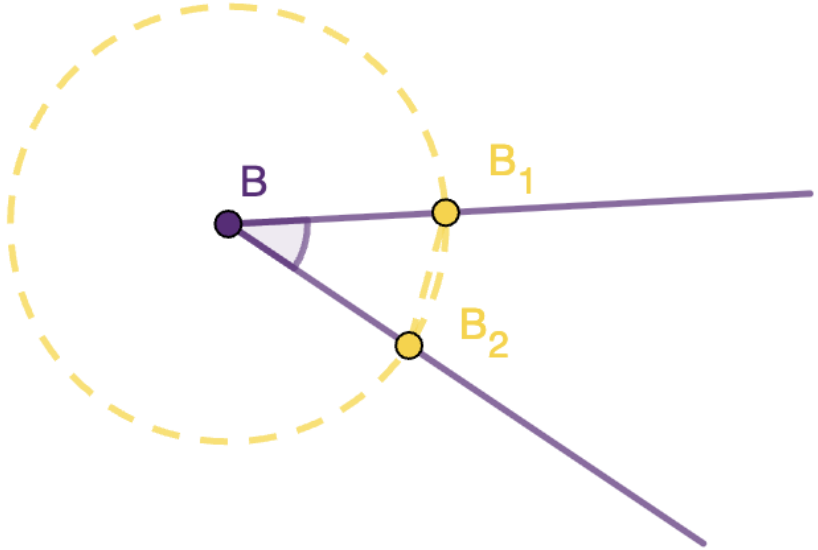

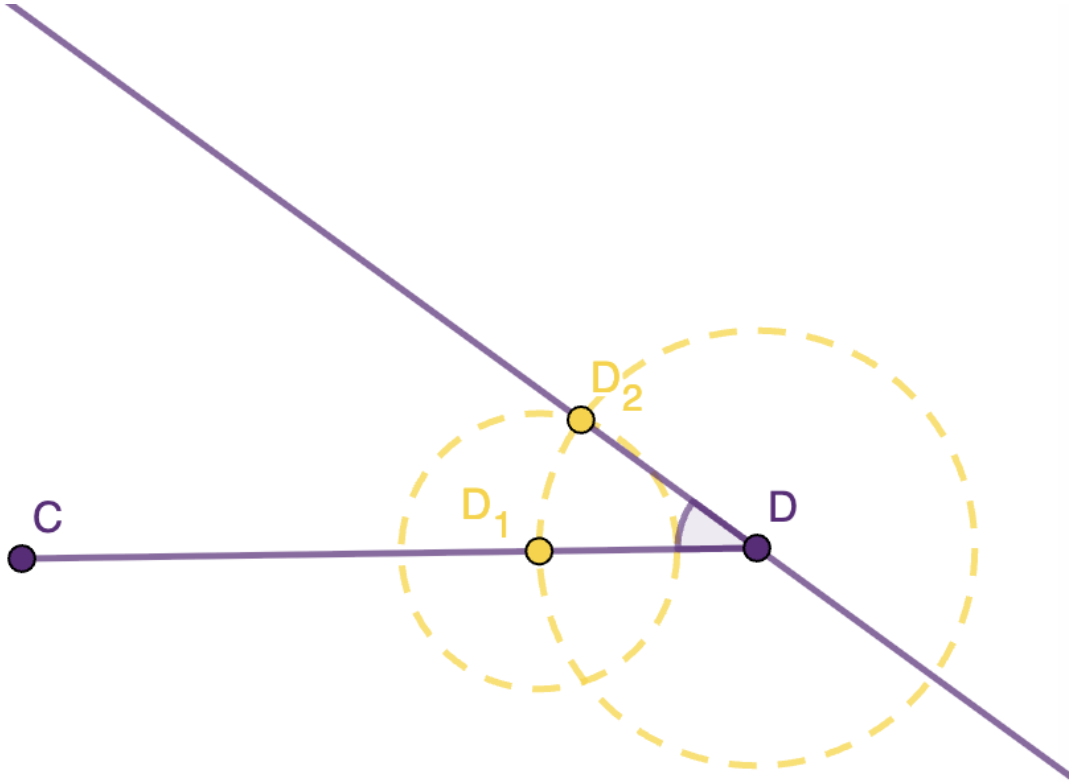

Building the toolset

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- Draw a segment on the line with a given length ✅
- Draw a perpendicular bisector ✅
- Find midpoint of given segment ✅
- Replicate given angle ✅
- Draw a parallel line

Building the toolset: parallel lines

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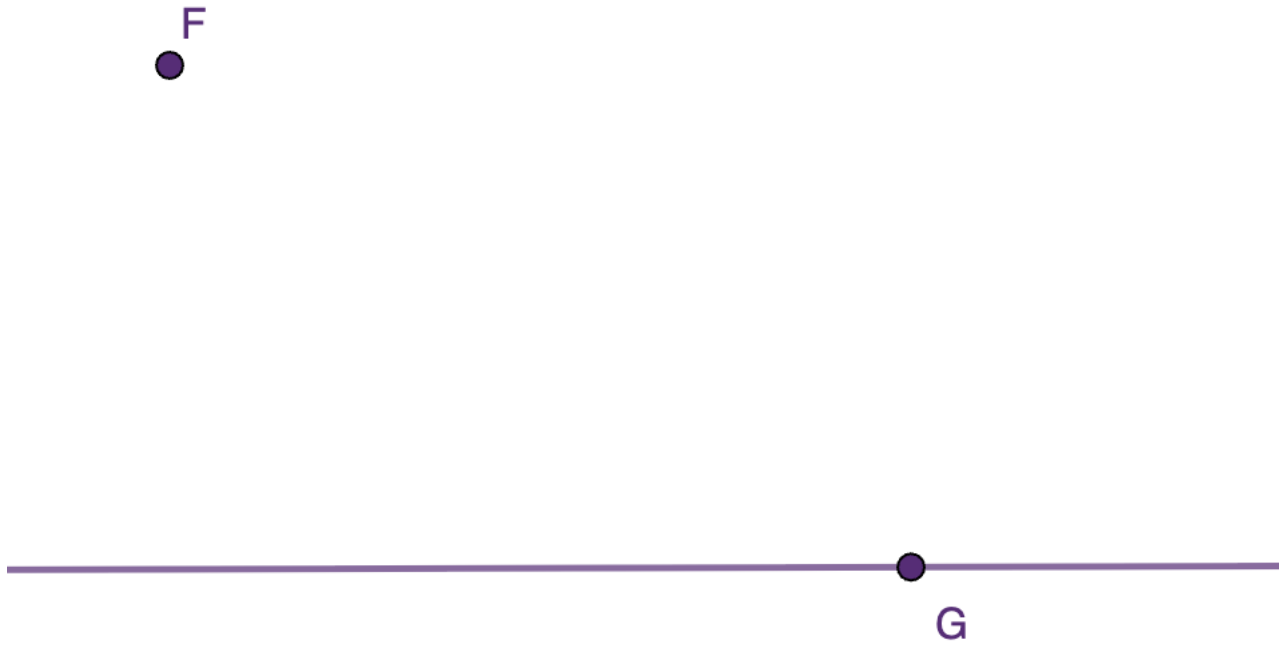

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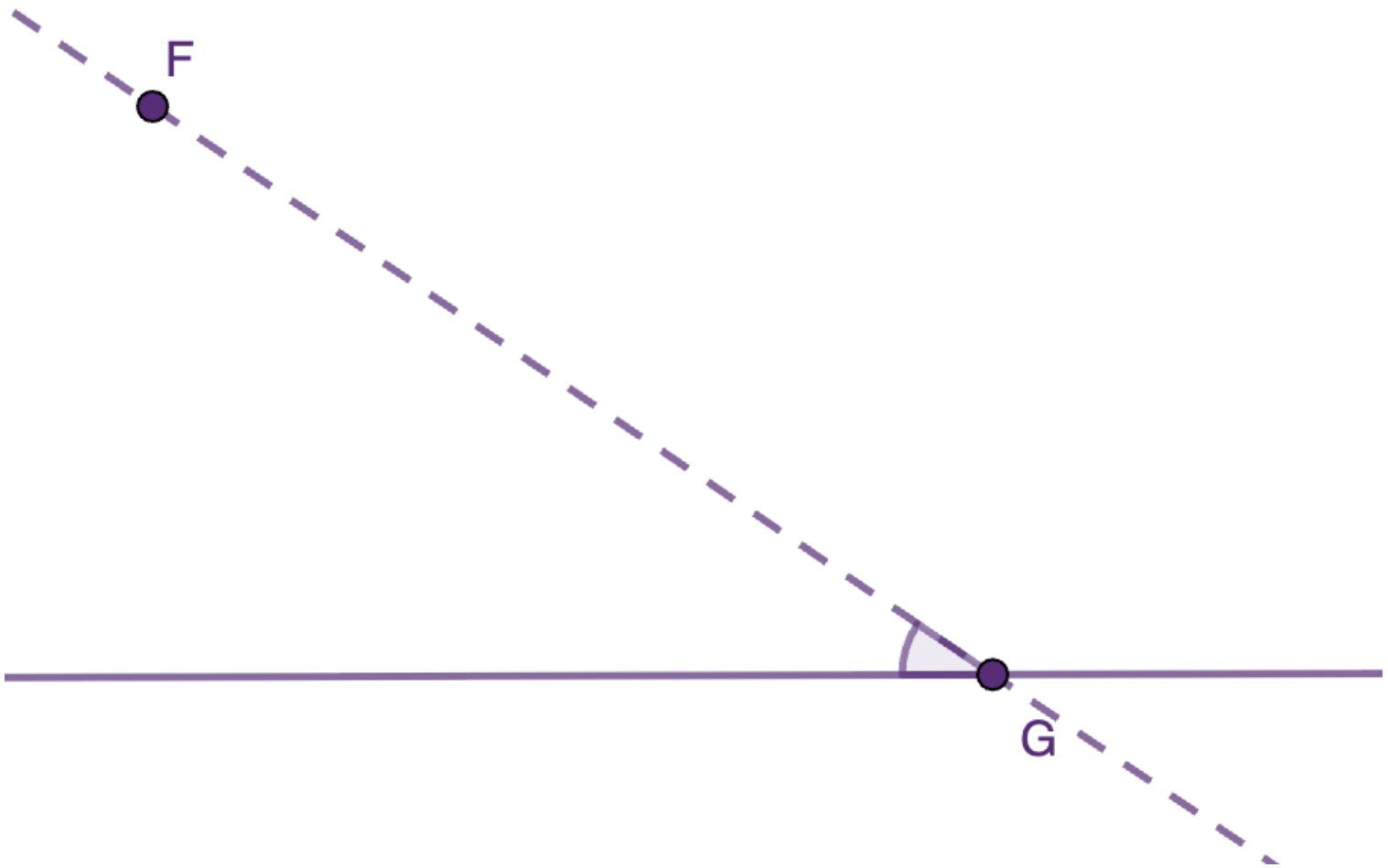

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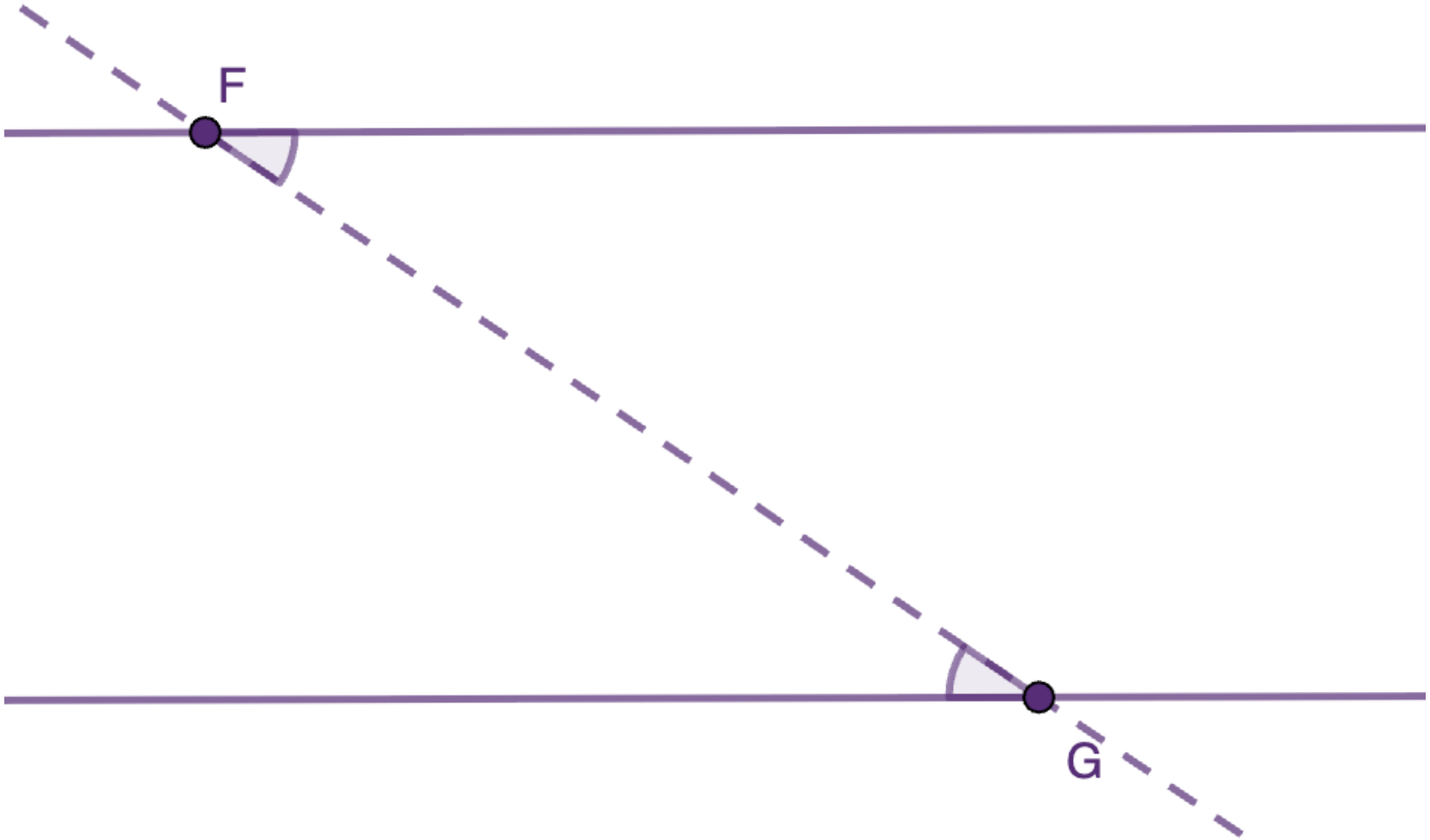

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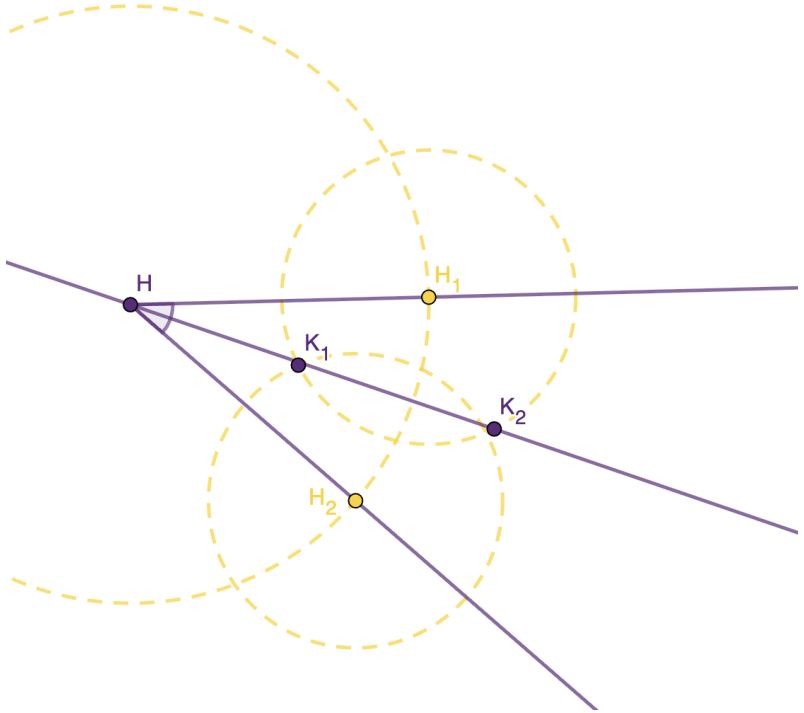
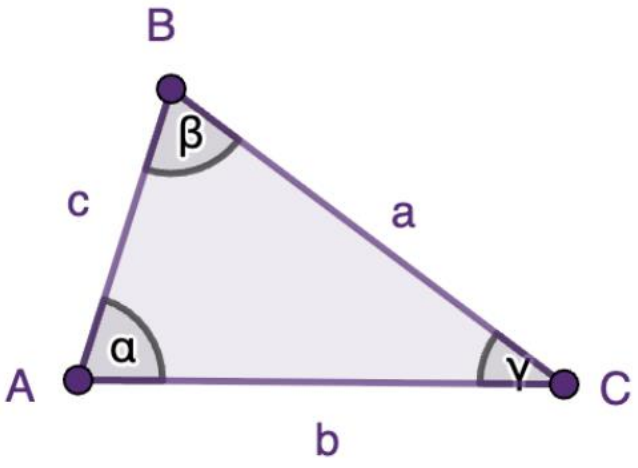
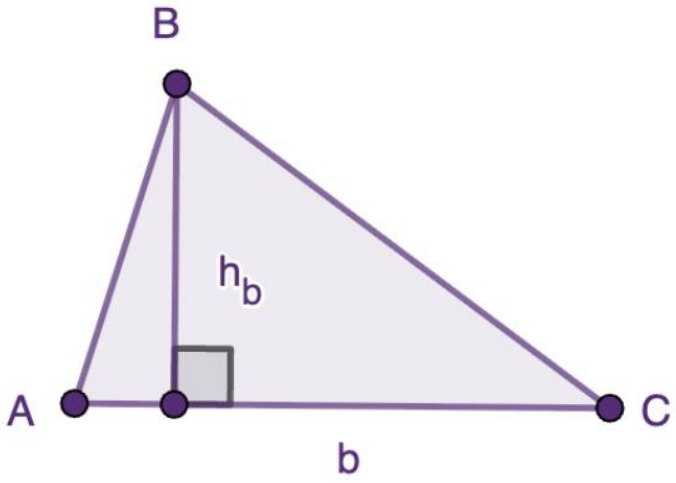


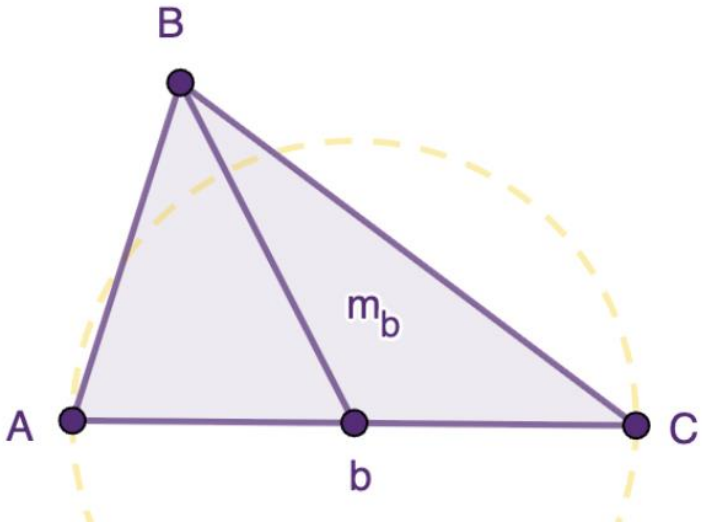
See further:

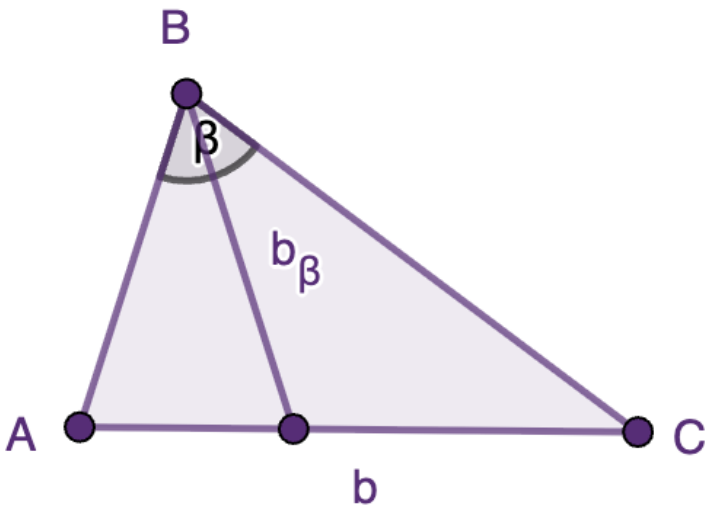
- Draw a bisector in a given angle
- Inscribed and circumscribed circles

Notation

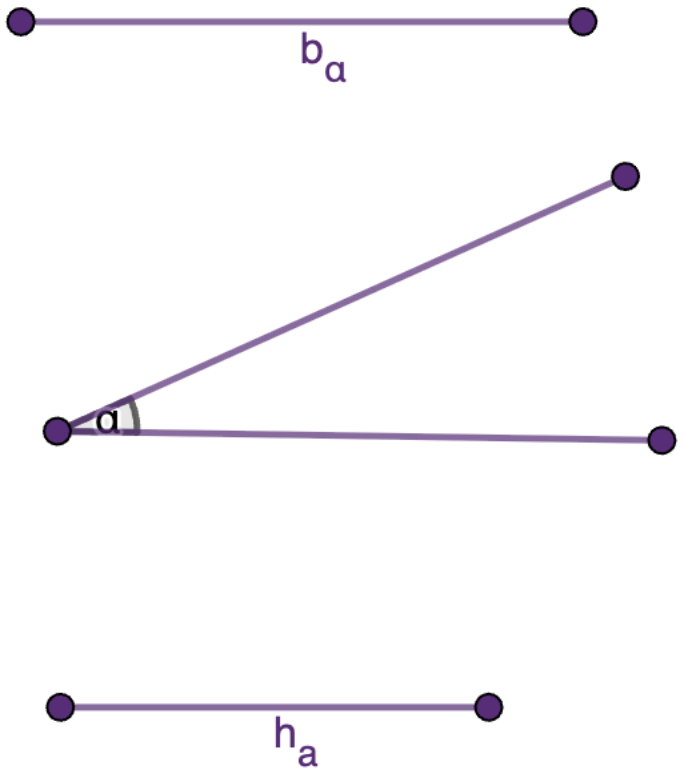






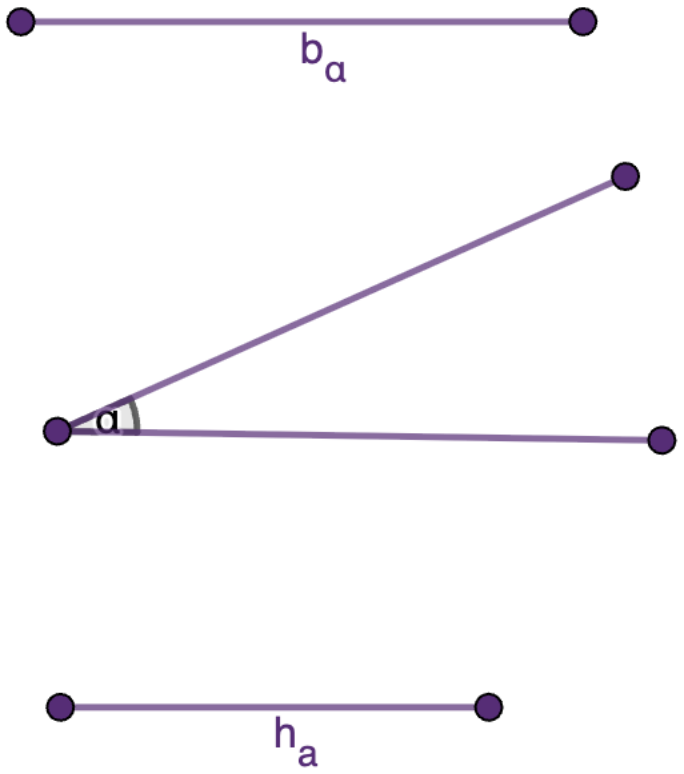


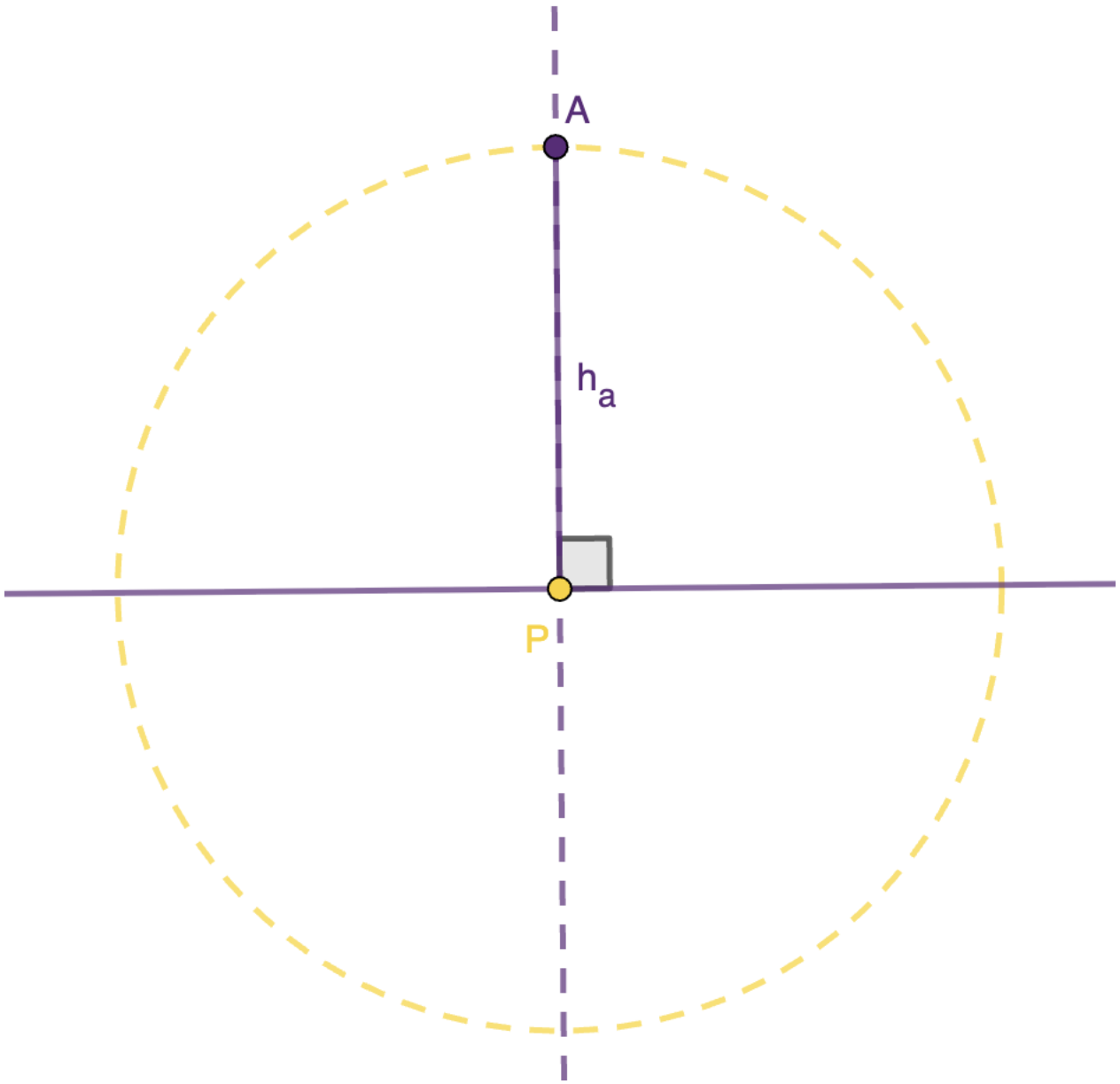
Answer to the question



- Can we reconstruct the triangle with one angle, one bisector and one of the medians?

Answer to the question





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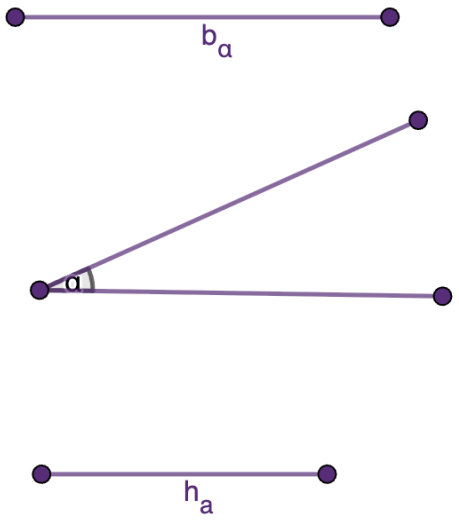

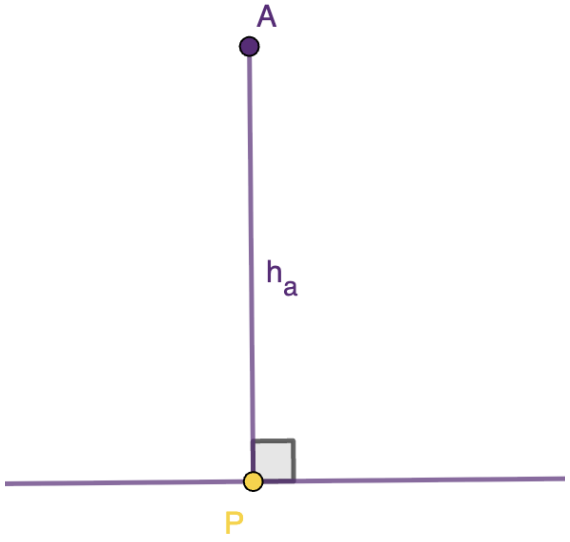

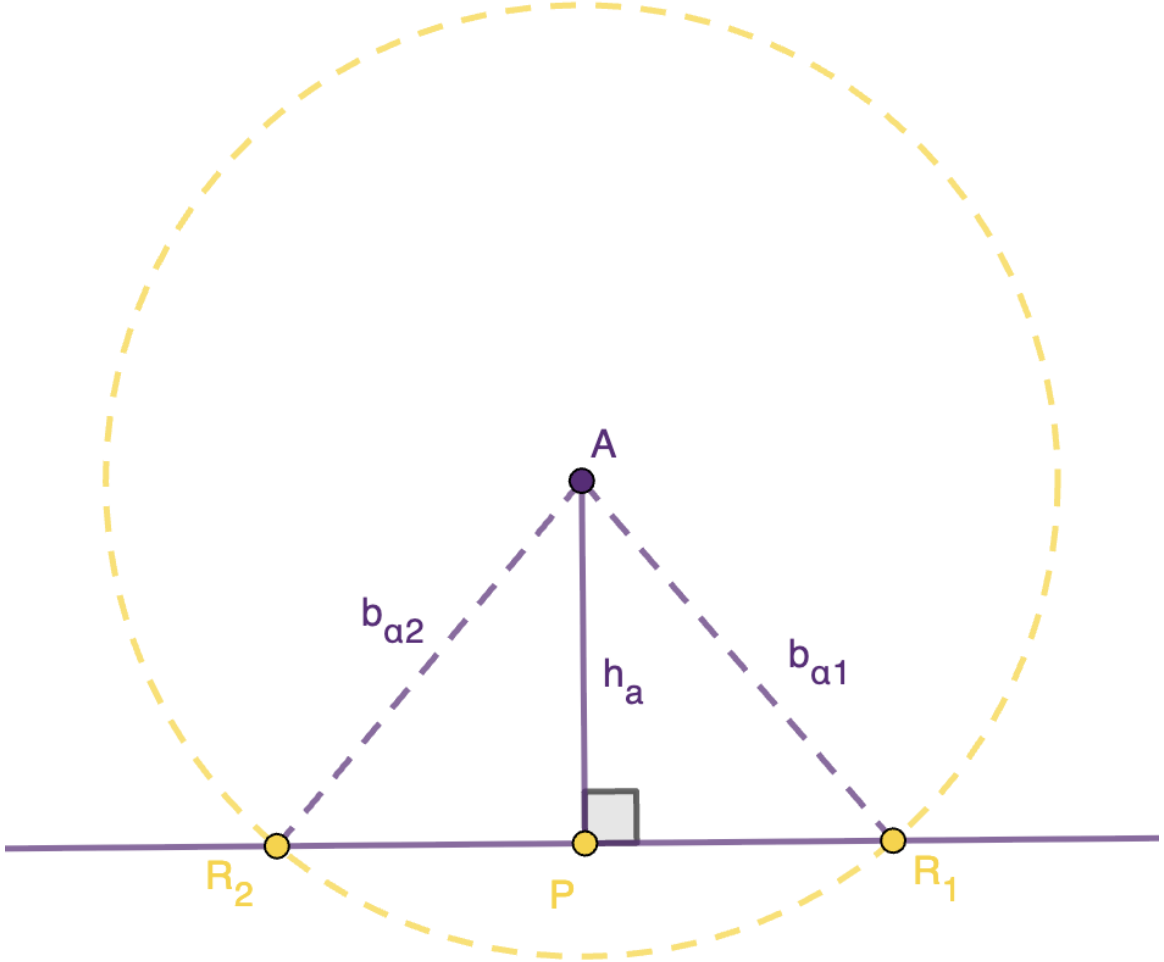

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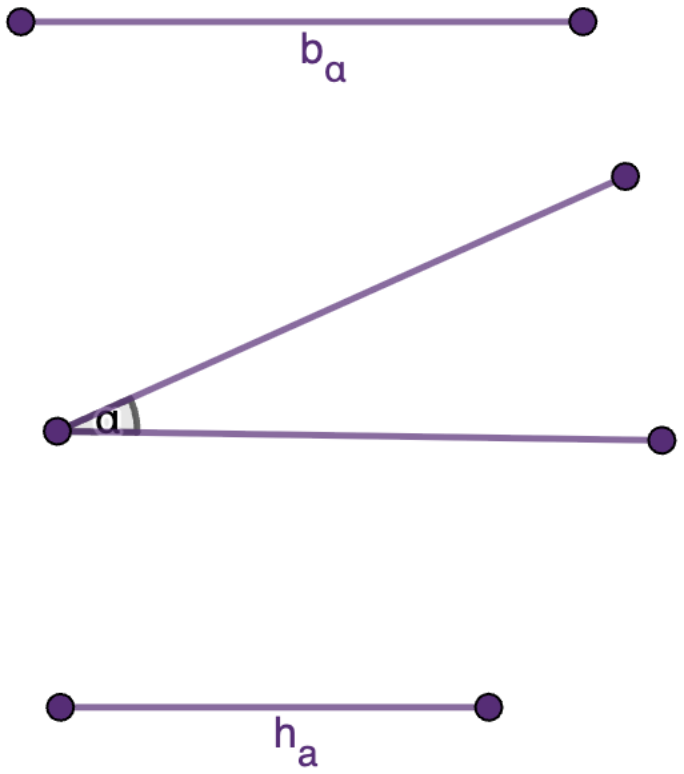

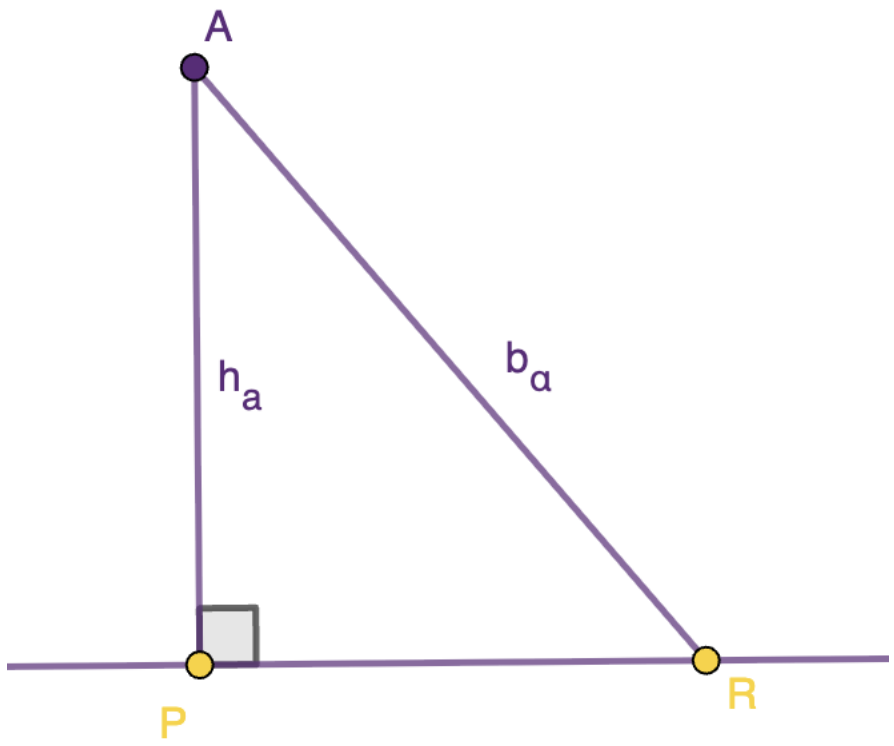

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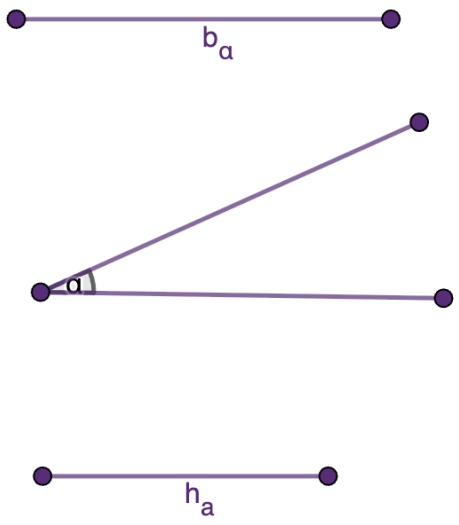

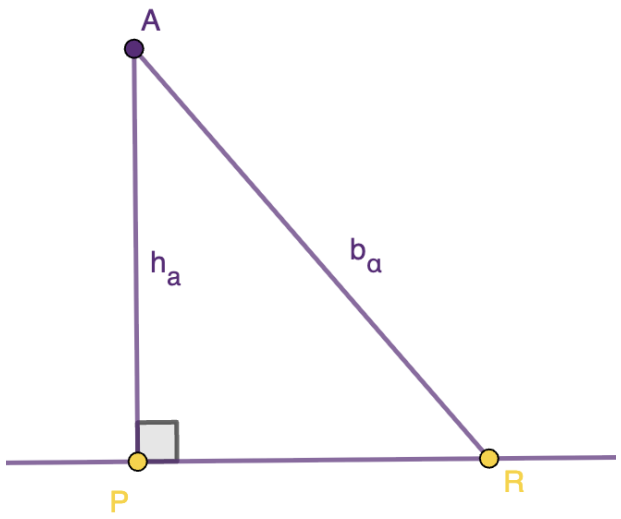

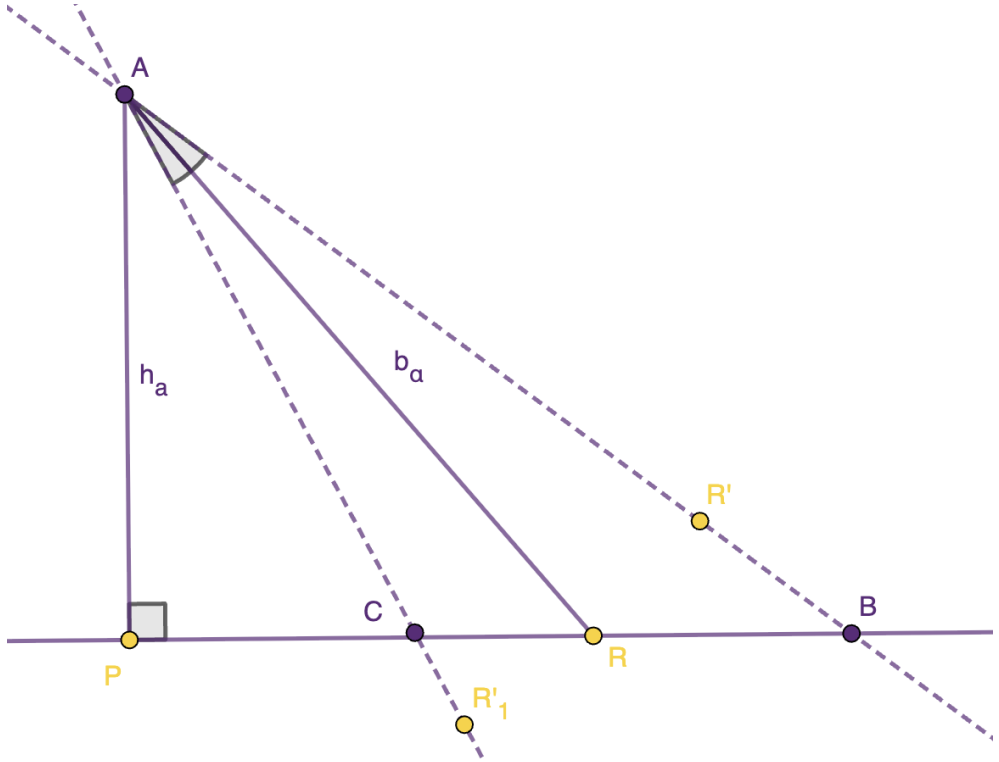

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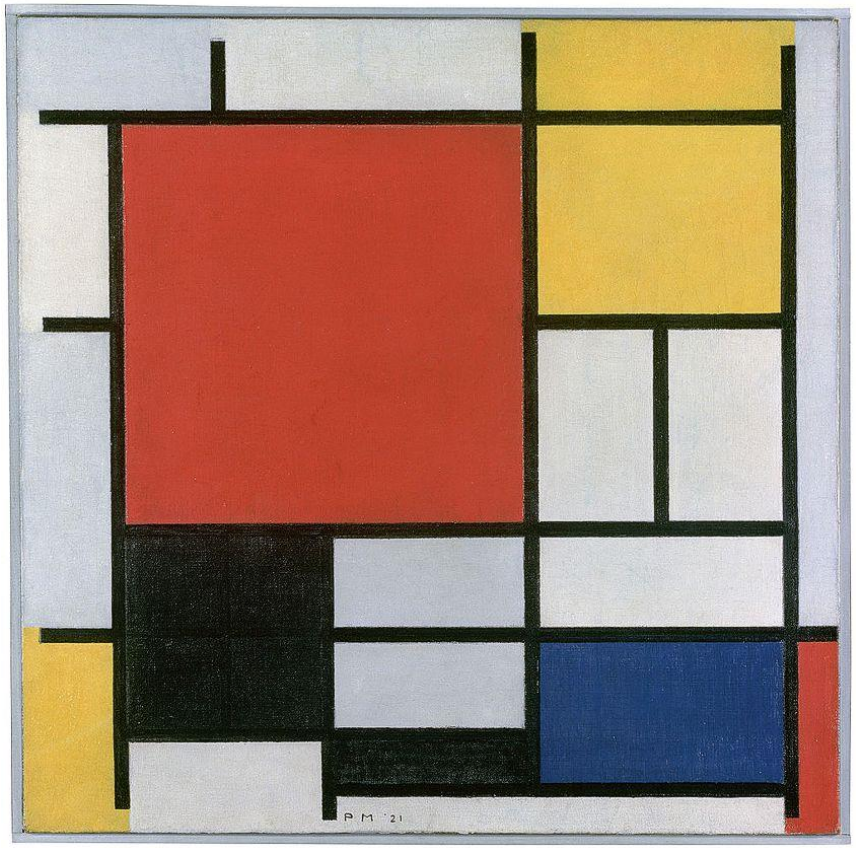


Source: https://artincontext.org/abstract-art/

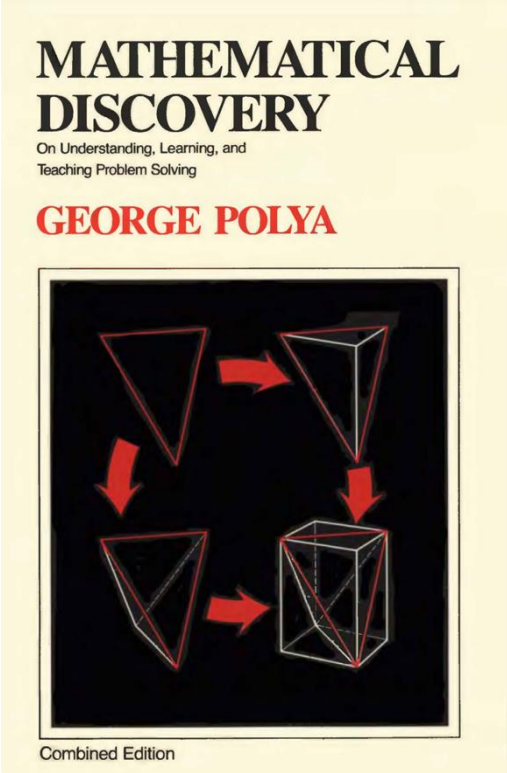
Homework

Given three points A, B, C.

Draw a line intersecting AC in the point X and BC in the point Y so that

$$AX = XY = YB$$

References



- Pólya, G. (1981a) Mathematical discovery: On understanding, learning, and teaching problem solving. New York: Wiley.
- Compass Equivalence theorem (2024) Wikipedia. Available at: https://en.wikipedia.org/wiki/Compass_equivalence_theorem (Accessed: 10 March 2025).
- Geogebra file: https://www.geogebra.org/calculator/pve2zqsu